

Bachelor of Information Technology

Database Systems

BIT1223/BTL1223/BCL1223

Chapter 5 Normalization

In this chapter, you will learn:

- What normalization is and what role it plays in the database design process
- About the normal forms 1NF, 2NF, 3NF
- How normal forms can be transformed from lower normal forms to higher normal forms
- That normalization and ER modeling are used concurrently to produce a good database design
- That some situations require denormalization to generate information efficiently

Data Anomalies

* recall from chapter 1

- An **anomaly** is an error or inconsistency that may occur when we try to update a table that contains redundant data.
- Types of data anomalies:
 - Update anomalies
 - Occur when changes must be made to existing records
 - Insertion anomalies
 - Occur when entering new records
 - Deletion anomalies
 - Occur when deleting records

Update Anomalies

Figure 1

EmpID	DeptID	EmpName	DeptName	Location
1	TRN	Jane	Training	Blk 31
2	TRN	Kim	Training	Blk 31
3	TRN	Peter	Training	Blk 31
4	HR	Mark	Human Resources	Blk 1
5	HR	Peter	Human Resources	Blk 1
6	FIN	Mel	Finance	Blk 1

EmpDept (EmpID, DeptID, EmpName, DeptName, Location)

- *Figure 1* shows a poorly-designed relation EmpDept which stores data of employees as well as data of departments they work in.
- Its primary key (PK) is EmpID.

1. Insertion Anomaly

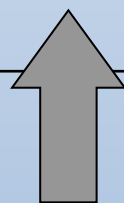
An **insertion anomaly** occurs when extra data besides the required data for insertion must be added to the database.

- There are two types of insertion anomaly which will be illustrated with two different scenarios.
- **1) First Type of Insertion Anomaly**
 - To insert the details of new employee into EmpDept relation, the details of the department which the employee is working must also be included.

Relation: EmpDept

EmpID	DeptID	EmpName	DeptName	Location
1	TRN	Jane	Training	Blk 31
2	TRN	Kim	Training	Blk 31
3	TRN	Peter	Training	Blk 31
4	HR	Mark	Human Resources	Blk 1
5	HR	Peter	Human Resources	Blk 1
6	FIN	Mel	Finance	Blk 1

New Employee: Mary



Department details must be consistent with existing records.

7	TRN	Mary	Training	Blk 31
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Figure 2 First type of Insertion Anomaly

- 2) 2nd Type of Insertion Anomaly

- To insert the details of a new department that has no employees into EmpDept relation, it is necessary to provide a value for EmpID, the primary key (PK).
 - A null value for the primary key is not allowed as it violates entity integrity.

Relation: EmpDept

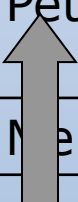
EmpID	DeptID	EmpName	DeptName	Location
1	TRN	Jane	Training	Blk 31
2	TRN	Kim	Training	Blk 31
3	TRN	Peter	Training	Blk 31
4	HR	Mark	Human Resources	Blk 1
5	HR	Peter		
6	FIN	Mel		
New Department: Sales  <i>New Department details cannot be inserted until an employee joins the department.</i>				
?	SAL	?	Sales	Blk 8

Figure 3 First type of Insertion Anomaly

2. Deletion Anomaly

A **deletion anomaly** occurs when deleting a row causes other data to be deleted.

- If we delete a tuple that represents the last employee working at a department from EmpDept relation, the details of the department will also disappear from the database.
- As shown in *Figure 4*, employee with EmpNo 6 is the last employee of Finance department.

Relation: EmpDept

EmpID	DeptID	EmpName	DeptName	Location
1	TRN	Jane	Training	Blk 31
2	TRN	Kim	Training	Blk 31
3	TRN	Peter	Training	Blk 31
4	HR	Mark	Human Resources	Blk 1
5	HR	Peter	Human Resources	Blk 1
6	FIN	Mel	Finance	Blk 1

Figure 4 Deletion Anomaly

When the last employee of Finance department is deleted, details of Finance department are also deleted.

3. Modification Anomaly

A **modification anomaly** occurs when it is necessary to change multiple rows to modify a single fact.

- If the value of any attributes of a department changes, the tuples of all employees working in that department must be updated.
- This situation of making multiple updates to ensure data consistency in the database can give rise to the problem of modification anomalies.

Relation: EmpDept

EmpID	DeptID	EmpName	DeptName	Location
1	TRN	Jane	Training	Blk 31
2	TRN	Kim	Training	Blk 31
3	TRN	Peter	Training	Blk 31
4	HR	Mark	Human Resources	Blk 1
5	HR	Peter	Human Resources	Blk 1
6	FIN	Mel	Finance	Blk 1

Figure 5 Modification Anomaly

If Training department changes its location from 'Blk 31' to 'Blk 27', then three tuples containing information of that department must be updated.

Database Tables and Normalization

- Normalization
 - Process for evaluating and correcting table structures to minimize data redundancies
 - Reduces data anomalies
 - Works through a series of stages called normal forms:
 - First normal form (1NF)
 - Second normal form (2NF)
 - Third normal form (3NF)
 - 2NF is better than 1NF; 3NF is better than 2NF
 - For most business database design purposes, 3NF is as high as we need to go in normalization process
 - Highest level of normalization is not always most desirable

The Need for Normalization

- Example: Company that manages building projects
 - Charges its clients by billing hours spent on each contract
 - Hourly billing rate is dependent on employee's position
 - Periodically, report is generated that contains information displayed below:

TABLE 5.1 A Sample Report Layout

PROJ. NUM.	PROJECT NAME	EMPLOYEE NUMBER	EMPLOYEE NAME	JOB CLASS.	CHG/ HOUR	HOURS BILLED	TOTAL CHARGE
15	Evergreen	103	June E. Arbough	Elec. Engineer	\$ 85.50	23.8	\$ 2,011.10
		101	John G. News	Database Designer	\$105.00	19.4	\$ 2,037.00
		105	Alice K. Johnson*	Database Designer	\$105.00	35.7	\$ 3,748.50
		106	William Smithfield	Programmer	\$ 35.75	12.6	\$ 450.45
		102	David H. Senior	Systems Analyst	\$ 96.75	23.8	\$ 2,302.65
				Subtotal			\$10,549.70
18	Amber Wave	114	Annelise Jones	Applications Designer	\$ 48.10	25.6	\$ 1,183.26
		118	James J. Frommer	General Support	\$ 18.36	45.3	\$ 831.71
		104	Anne K. Ramoras*	Systems Analyst	\$ 96.75	32.4	\$ 3,135.70
		112	Darlene M. Smithson	DSS Analyst	\$ 45.95	45.0	\$ 2,021.80
				Subtotal			\$ 7,172.47
22	Rolling Tide	105	Alice K. Johnson	Database Designer	\$105.00	65.7	\$ 6,793.50
		104	Anne K. Ramoras	Systems Analyst	\$ 96.75	48.4	\$ 4,682.70
		113	Delbert K. Joenbrood*	Applications Designer	\$ 48.10	23.6	\$ 1,135.16
		111	Geoff B. Wabash	Clerical Support	\$ 26.87	22.0	\$ 591.14
		106	William Smithfield	Programmer	\$ 35.75	12.8	\$ 457.60
				Subtotal			\$13,660.10
25	Starflight	107	Maria D. Alonzo	Programmer	\$ 35.75	25.6	\$ 879.45
		115	Travis B. Bawangi	Systems Analyst	\$ 96.75	45.8	\$ 4,431.15
		101	John G. News*	Database Designer	\$105.00	56.3	\$ 5,911.50
		114	Annelise Jones	Applications Designer	\$ 48.10	33.1	\$ 1,592.11
		108	Ralph B. Washington	Systems Analyst	\$ 96.75	23.6	\$ 2,283.30
		118	James J. Frommer	General Support	\$ 18.36	30.5	\$ 559.98
		112	Darlene M. Smithson	DSS Analyst	\$ 45.95	41.4	\$ 1,902.33
				Subtotal			\$17,559.82
				Total			\$48,942.09

Note: * indicates project leader.

The Need for Normalization (continued)

- Structure of data set in Table 5.1 does not handle data very well
- The table structure appears to work; report generated with ease
- Unfortunately, report may yield different results depending on what data anomaly has occurred
- Each table represents a single subject
- No data item will be unnecessarily stored in more than one table
- All attributes in a table are dependent on the primary key

The Normalization Process

- Each table represents a single subject
- No data item will be unnecessarily stored in more than one table
- All attributes in a table are dependent on the primary key

TABLE 5.2 Normal Forms		
NORMAL FORM	CHARACTERISTIC	SECTION
First normal form (1NF)	Table format; no repeating groups and PK identified	5.3.1
Second normal form (2NF)	1NF and no partial dependencies	5.3.2
Third normal form (3NF)	2NF and no transitive dependencies	5.3.3
Boyce-Codd normal form (BCNF)	Every determinant is a candidate key (special case of 3NF)	5.6.1
Fourth normal form (4NF)	3NF and no independent multivalued dependencies	5.6.2

Conversion to First Normal Form

- Repeating group
 - Derives its name from the fact that a group of multiple entries of same type can exist for any single key attribute occurrence
- Relational table must not contain repeating groups
- Normalizing table structure will reduce data redundancies
- Normalization is three-step procedure
- **Step 1: Eliminate the Repeating Groups**
 - Present data in tabular format, where each cell has single value and there are no repeating groups
 - Eliminate repeating groups, eliminate nulls by making sure that each repeating group attribute contains an appropriate data value

Conversion to First Normal Form (continued)

**FIGURE
5.2**

A table in first normal form

Table name: DATA_ORG_1NF

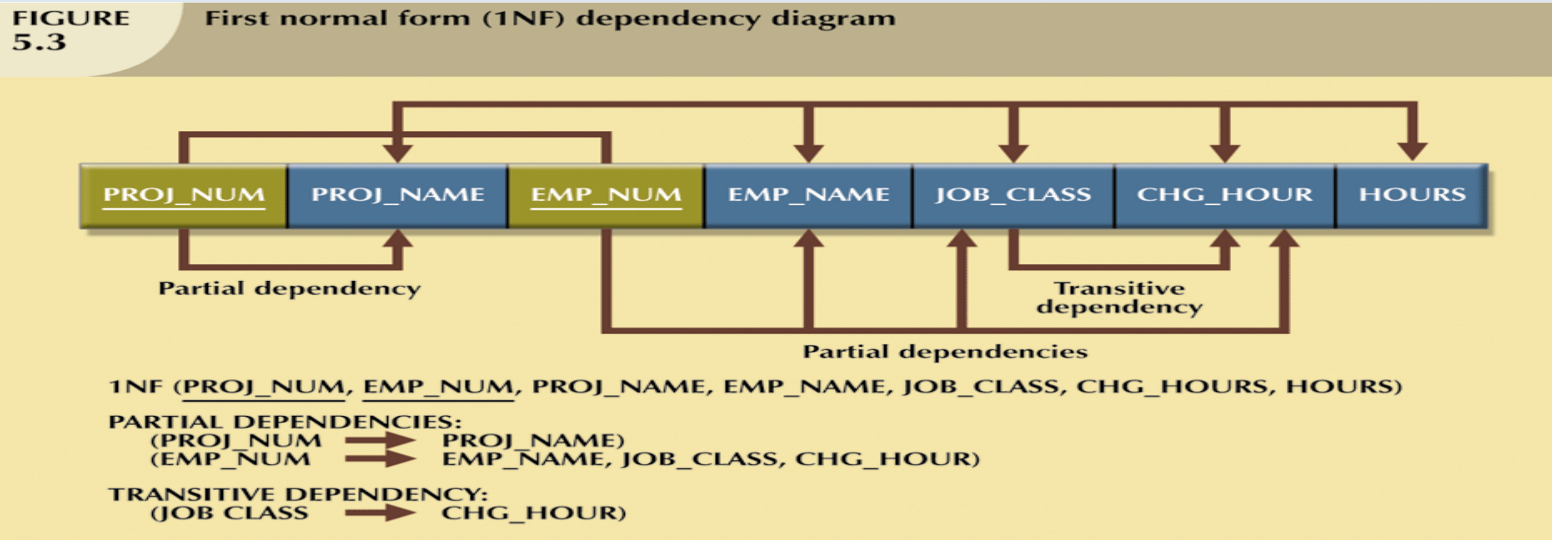
Database name: Ch05_ConstructCo

	PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
▶	15	Evergreen	103	June E. Arbough	Elect. Engineer	\$84.50	23.8
	15	Evergreen	101	John G. News	Database Designer	\$105.00	19.4
	15	Evergreen	105	Alice K. Johnson *	Database Designer	\$105.00	35.7
	15	Evergreen	106	William Smithfield	Programmer	\$35.75	12.6
	15	Evergreen	102	David H. Senior	Systems Analyst	\$96.75	23.8
	18	Amber Wave	114	Annelise Jones	Applications Designer	\$48.10	24.6
	18	Amber Wave	118	James J. Frommer	General Support	\$18.36	45.3
	18	Amber Wave	104	Anne K. Ramoras *	Systems Analyst	\$96.75	32.4
	18	Amber Wave	112	Darlene M. Smithson	DSS Analyst	\$45.95	44.0
	22	Rolling Tide	105	Alice K. Johnson	Database Designer	\$105.00	64.7
	22	Rolling Tide	104	Anne K. Ramoras	Systems Analyst	\$96.75	48.4
	22	Rolling Tide	113	Delbert K. Joenbrood *	Applications Designer	\$48.10	23.6
	22	Rolling Tide	111	Geoff B. Wabash	Clerical Support	\$26.87	22.0
	22	Rolling Tide	106	William Smithfield	Programmer	\$35.75	12.8
	25	Starflight	107	Maria D. Alonzo	Programmer	\$35.75	24.6
	25	Starflight	115	Travis B. Bawangi	Systems Analyst	\$96.75	45.8
	25	Starflight	101	John G. News *	Database Designer	\$105.00	56.3
	25	Starflight	114	Annelise Jones	Applications Designer	\$48.10	33.1
	25	Starflight	108	Ralph B. Washington	Systems Analyst	\$96.75	23.6
	25	Starflight	118	James J. Frommer	General Support	\$18.36	30.5
	25	Starflight	112	Darlene M. Smithson	DSS Analyst	\$45.95	41.4

Conversion to First Normal Form (continued)

- **Step 2: Identify the Primary Key**
 - Primary key must uniquely identify attribute value
 - New key must be composed
- **Step 3: Identify All Dependencies**
 - Dependencies can be depicted with help of a diagram
 - Dependency diagram:
 - Depicts all dependencies found within given table structure
 - Helpful in getting bird's-eye view of all relationships among table's attributes
 - Makes it less likely that will overlook an important dependency

Conversion to First Normal Form (continued)



- Some tables contain partial dependencies
 - Dependencies based on only part of the primary key
 - Sometimes used for performance reasons, but should be used with caution
 - Still subject to data redundancies

Conversion to Second Normal Form

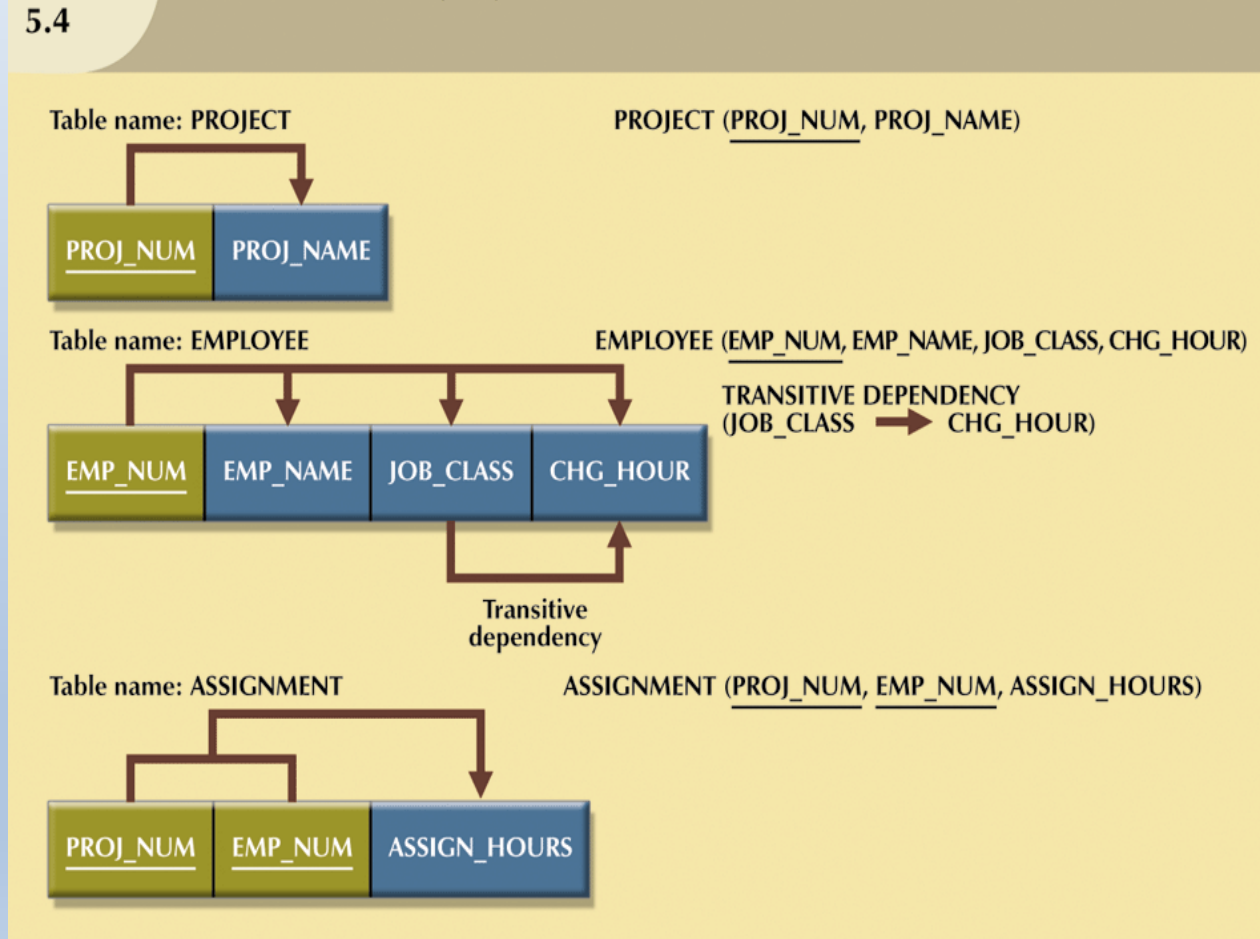
- Relational database design can be improved by converting the database into second normal form (2NF)
- Two steps
- **Step 1: Write Each Key Component on a Separate Line**
 - Write each key component on separate line, then write original (composite) key on last line
 - Each component will become key in new table
- **Step 2: Assign Corresponding Dependent Attributes**
 - Determine those attributes that are dependent on other attributes
 - At this point, most anomalies have been eliminated

Conversion to Second Normal Form (continued)

- Table is in second normal form (2NF) when:
 - It is in 1NF and
 - It includes no partial dependencies:
 - No attribute is dependent on only portion of primary key

FIGURE 5.4

Second normal form (2NF) conversion results



Conversion to Third Normal Form

- Data anomalies created are easily eliminated by completing three steps
- **Step 1: Identify Each New Determinant**
 - For every transitive dependency, write its determinant as PK for new table
 - Determinant
 - Any attribute whose value determines other values within a row
- **Step 2: Identify the Dependent Attributes**
 - Identify attributes dependent on each determinant identified in Step 1 and identify dependency
 - Name table to reflect its contents and function

Conversion to Third Normal Form (continued)

- **Step 3: Remove the Dependent Attributes from Transitive Dependencies**
 - Eliminate all dependent attributes in transitive relationship(s) from each of the tables that have such a transitive relationship
 - Draw new dependency diagram to show all tables defined in Steps 1–3
 - Check new tables as well as tables modified in Step 3 to make sure that each table has determinant and that no table contains inappropriate dependencies

Conversion to Third Normal Form (cont)

FIGURE 5.5

Third normal form (3NF) conversion results

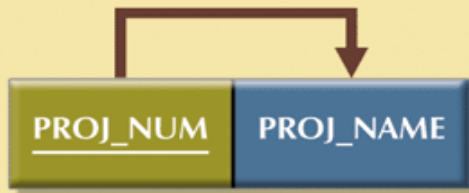


Table name: PROJECT

PROJECT (PROJ_NUM, PROJ_NAME)



Table name: JOB

JOB (JOB_CLASS, CHG_HOUR)

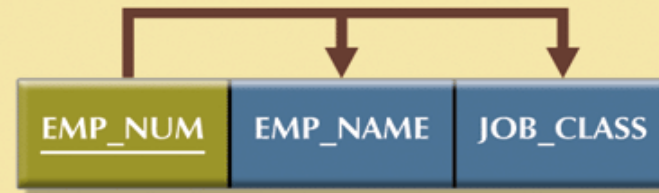


Table name: EMPLOYEE

EMPLOYEE (EMP_NUM, EMP_NAME, JOB_CLASS)

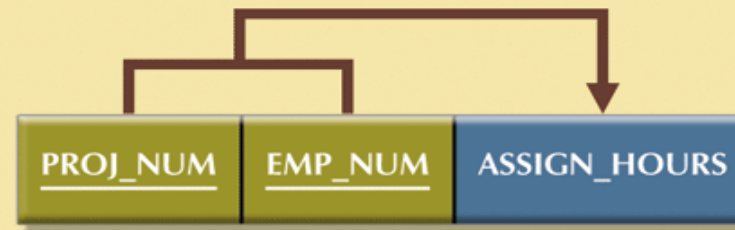


Table name: ASSIGNMENT

ASSIGNMENT (PROJ_NUM, EMP_NUM, ASSIGN_HOURS)

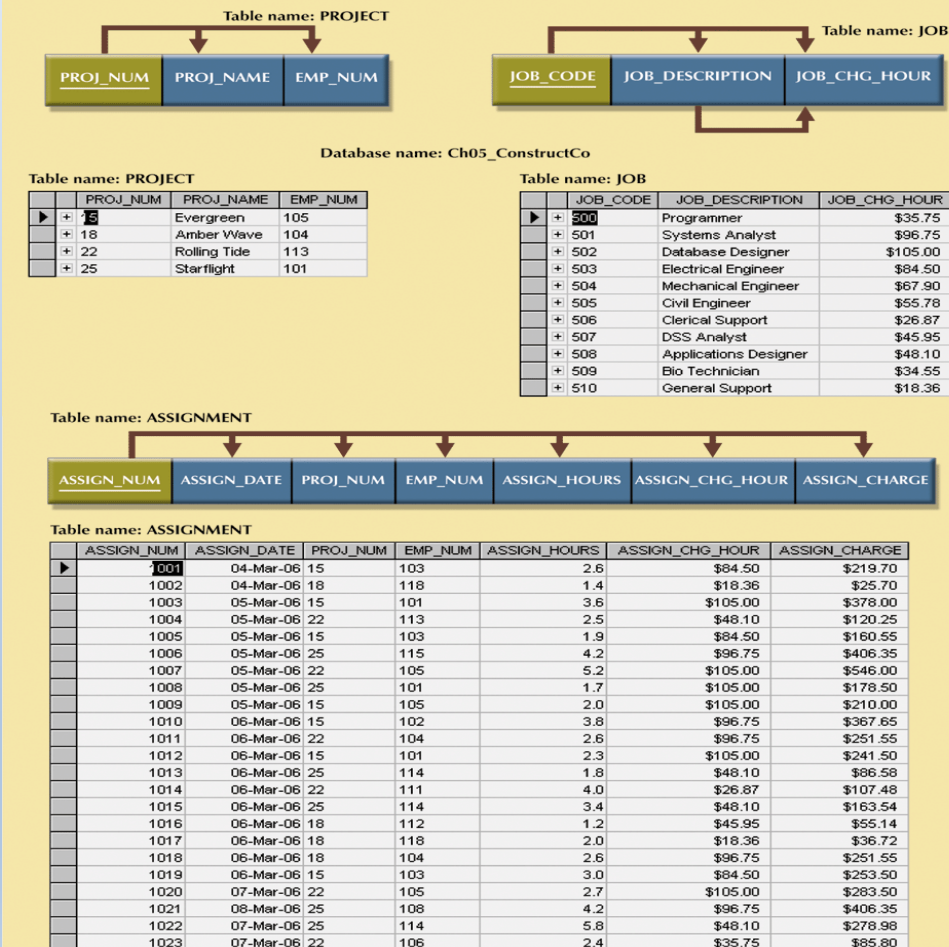
- A table is in third normal form (3NF) when both of the following are true:
 - It is in 2NF
 - It contains no transitive dependencies

Improving the Design

- Table structures are cleaned up to eliminate troublesome initial partial and transitive dependencies
- Normalization cannot, by itself, be relied on to make good designs
- It is valuable because its use helps eliminate data redundancies
- Issues to address in order to produce a good normalized set of tables:
 - Evaluate PK Assignments
 - Evaluate Naming Conventions
 - Refine Attribute Atomicity
 - Identify New Attributes
 - Identify New Relationships
 - Refine Primary Keys as Required for Data Granularity
 - Maintain Historical Accuracy
 - Evaluate Using Derived Attributes

Improving the Design (continued)

FIGURE 5.6 The completed database



The completed database (continued)

Table name: EMPLOYEE

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_HIREDATE	JOB_CODE
101	News	John	G	08-Nov-00	502
102	Senior	David	H	12-Jul-89	501
103	Arbough	June	E	01-Dec-97	503
104	Ramoras	Anne	K	15-Nov-88	501
105	Johnson	Alice	K	01-Feb-94	502
106	Smithfield	William		22-Jun-05	500
107	Alonzo	Maria	D	10-Oct-94	500
108	Washington	Ralph	B	22-Aug-89	501
109	Smith	Larry	W	18-Jul-99	501
110	Olenko	Gerald	A	11-Dec-96	505
111	Wabash	Geoff	B	04-Apr-89	506
112	Smithson	Darlene	M	23-Oct-95	507
113	Joebrood	Delbert	K	15-Nov-94	508
114	Jones	Annelise		20-Aug-91	508
115	Bawangi	Travis	B	25-Jan-90	501
116	Pratt	Gerald	L	05-Mar-95	510
117	Williamson	Angie	H	19-Jun-94	509
118	Frommer	James	J	04-Jan-06	510

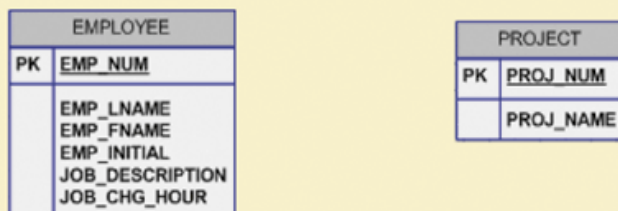
Normalization and Database Design

- Normalization should be part of design process
- Make sure that proposed entities meet required normal form before table structures are created
- Many real-world databases have been improperly designed or burdened with anomalies if improperly modified during course of time
- You may be asked to redesign and modify existing databases
- ER diagram
 - Provides big picture, or macro view, of an organization's data requirements and operations
 - Created through an iterative process
 - Identifying relevant entities, their attributes and their relationship
 - Use results to identify additional entities and attributes

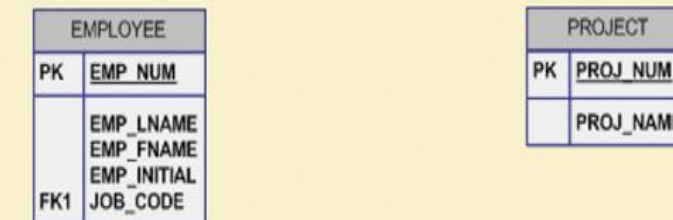
Normalization and Database Design (continued)

- Normalization procedures
 - Focus on characteristics of specific entities
 - Represents micro view of entities within ER diagram
- Difficult to separate normalization process from ER modeling process
- Two techniques should be used concurrently

FIGURE 5.12 Initial contracting company ERD



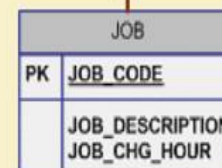
Modified contracting company ERD



is held by

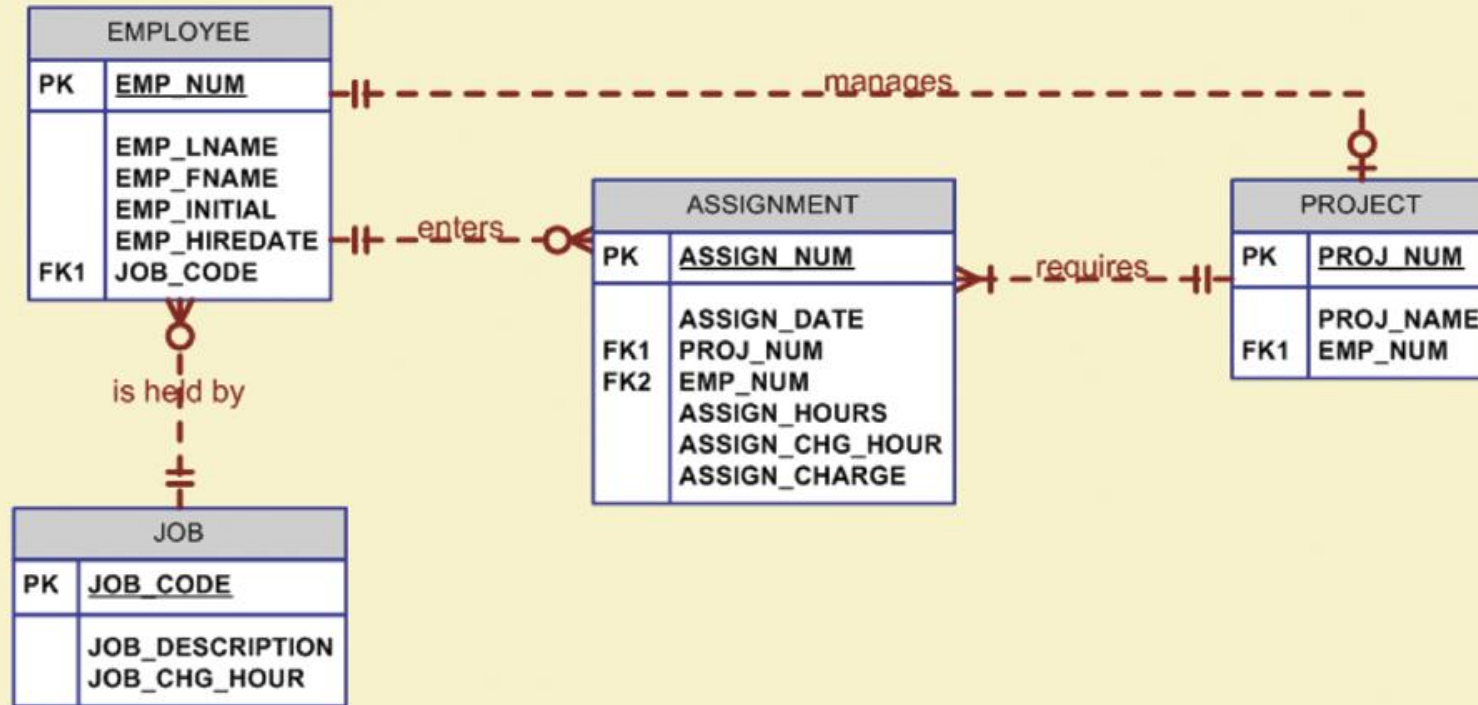
Each EMPLOYEE has one (main) JOB classification.
Any JOB classification may be held by many EMPLOYEES.

Some JOB classifications have not yet been staffed.
Therefore, EMPLOYEE is optional to JOB.



Normalization and Database Design (continued)

FIGURE 5.15 Final contracting company ERD



Denormalization

- Creation of normalized relations is important database design goal
- Processing requirements should also be a goal
- If tables decomposed to conform to normalization requirements:
 - Number of database tables expands
- Joining the larger number of tables takes additional input/output (I/O) operations and processing logic, thereby reducing system speed
- Conflicts between design efficiency, information requirements, and processing speed are often resolved through compromises that may include denormalization

Denormalization (continued)

- Unnormalized tables in production database tend to suffer from these defects:
 - Data updates are less efficient because programs that read and update tables must deal with larger tables
 - Indexing is more cumbersome
 - Unnormalized tables yield no simple strategies for creating virtual tables known as views
- Use denormalization cautiously
- Understand why—under some circumstances—unnormalized tables are better choice